PIE CHART, DONUT CHART, TREE MAP, MAP VISUALS AND GAUGE CHART IN POWER BI

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1. Introduction to One Quantitative and One Categorical Variable Visualization

In Power BI, several visuals are designed to represent the relationship between **one quantitative variable** (such as sales, profit, cost) and **one categorical variable** (such as region, city, sub-category). While bar charts and column charts are very commonly used for this purpose, Power BI includes additional templates such as:

- **Pie Chart**
- **Donut Chart**
- **Tree Map**
- **Map and Filled Map**
- **ArcGIS and Azure Maps**
- **Gauge Chart**

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Each of these visuals is appropriate for specific scenarios, depending on the number of categories, the type of analysis required, and the need for spatial or KPI-based representation.

The following sections explain each visual in depth, including when to use them, how they behave, and the formatting and analytical options available in Power BI.

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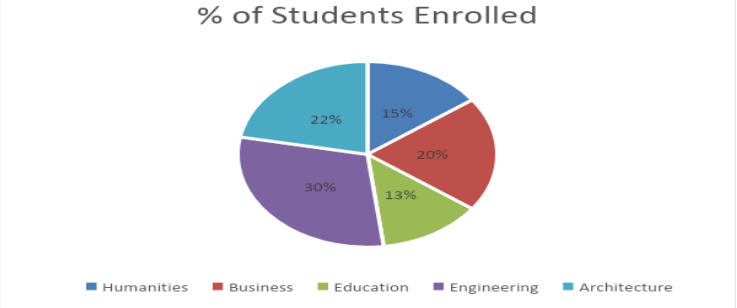
2. Pie Chart

2.1 Purpose and Use Case

A **Pie Chart** is one of the simplest methods to visualize the **distribution of a single quantitative variable across a limited number of categories**. It visually represents how much each category contributes to the overall total by dividing the circle into proportional slices.

Ideal Situations

- When the number of categories is **small (preferably up to 6–8 categories)**.
- When the goal is to **show proportional contribution** of categories to the whole.
- When comparison is simple and not highly detailed.



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Not Recommended When

- The number of categories exceeds 8.
- Distinction between slices becomes difficult.
- Values are very close, making visual distinction unclear.

2.2 Creating a Pie Chart in Power BI

Step-by-Step

1. Click on the **Pie Chart** visual in the Visualizations pane.
2. Expand the visual size so slices are clearly visible.
3. Drag a **quantitative variable** (e.g., *Sales*) into the **Values** field.
    - Power BI creates a full circle representing the total magnitude.
4. Drag a **categorical variable** (e.g., *Region*) into the **Legend** field.
    - The chart divides into slices for each category with distinct colors.

The two key fields are:

- **Values** → Quantitative measure
- **Legend** → Categorical variable

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2.3 Formatting Pie Chart

Using the **Format Your Visual** pane, you can customize:

a. Legend

- Position (top, bottom, left, right)
- Font size and color
- Category names (rename if needed)

b. Slices

- Change individual slice colors
- Add borders
- Adjust transparency or visibility

c. Rotation

- Modify rotation angle from the starting point (default 0°)
- Used when you want a specific category to appear at the top or left

d. Labels

- Show category names
- Show data values / percentages / both
- Adjust text size, display units, and color

e. Title, Background, and Effects

- Customize title text, font size, and alignment
- Add background color
- Add shadows and visual borders

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f. Analytics

Pie charts **do not support analytics lines** (like median, reference line) because they are circular rather than axis-based.

2.4 Drill Down in Pie Chart

Pie charts support **hierarchical drilling**:

- Add multiple categorical fields to the Legend section (or use category hierarchy).
- Enable **Drill Down** to move from high-level categories (like Region) to lower ones (Category → Subcategory).
- The user can drill up to return to the previous level.

This allows progressive exploration of categorical structures within the pie chart.

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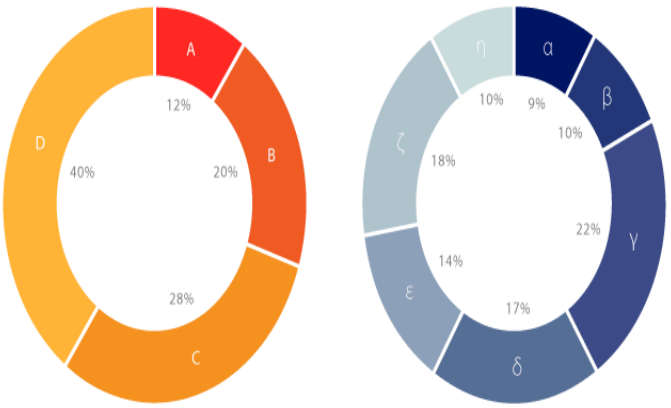
3. Donut Chart

3.1 Purpose and Characteristics

A **Donut Chart** is similar to a Pie Chart but contains a **hole in the center**, giving it a modern, cleaner look.

Why Use Donut Chart

- More contemporary visual design
- Center space can be used to place key KPIs (not default in Power BI but possible via custom visuals)
- Often easier to read than a solid pie



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Functionally, **it is identical to the pie chart** in terms of:

- Fields (Legend and Values)
- Formatting options
- Drill-down behavior

3.2 Creating Donut Chart

Just click the **Donut Chart** template and provide:

- **Values** → Sales
- **Legend** → Region

All formatting options remain similar to those of the pie chart.

3.3 When Donut Chart Is Not Suitable

Just like pie charts, when:

- Number of categories exceeds 8
- Visual becomes cluttered
- Colors become indistinguishable

In such scenarios, a **tree map** is a better option.

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4. Tree Map

4.1 Purpose and Use Case

A **Tree Map** is ideal for representing:

- A single quantitative variable
- With **many categories** (more than 8)

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Tree maps allocate space proportional to category magnitude in the form of rectangles, making it easier to interpret when many categories exist.

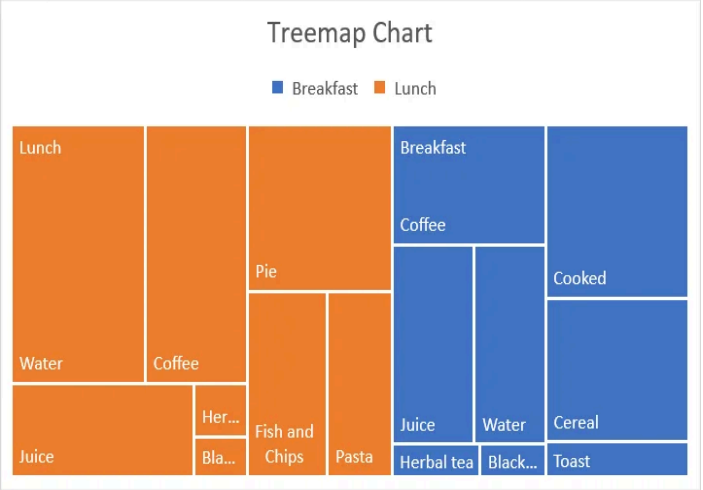
4.2 When to Use Tree Map

- When pie/donut charts are too cluttered
- When categories exceed 10–15
- When categories differ significantly in size
- When hierarchical drill-down is required

4.3 Creating Tree Map

Steps:

1. Click on the **Tree Map** visual template.
2. Provide:
    - **Values** → Sales
    - **Group** → City, Region, Category (any categorical variable)



Tree map displays:

- Large rectangles for categories with high values
- Smaller ones for smaller categories

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4.4 Drill Down in Tree Map

Tree maps support multiple categorical levels:

- Region → Category → Subcategory → City
- Drill Down and Drill Up buttons allow hierarchical exploration

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4.5 Advantages Over Pie/Donut Chart

- Handles dozens of categories clearly
- Shows relative magnitude visually through area
- Simple to compare large vs. small categories

5. Map Visuals in Power BI

Power BI provides multiple map visual templates:

a. Map (Bubble Map)

Shows data points as circles at geographic locations.

b. Filled Map

Colors regions (such as states, countries) based on a measure.

c. ArcGIS for Power BI

Provides advanced mapping capabilities (requires internet).

d. Azure Maps

Requires login; often limited in demonstration scenarios.

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5.1 Standard Map (Bubble Map)

5.1.1 Use Case

Useful when:

- You want to show geographic distribution of data
- Locations are available as:
    - Country
    - State
    - City
    - Latitude/Longitude

5.1.2 Creating the Map

1. Click on the **Map** template.
2. Provide:
    - **Location** → State/City
    - **Values** → Sales or Profit

Power BI attempts intelligent geocoding:

- Using state names, it finds exact map positions.
- Bubbles represent magnitude (larger = higher values).

5.1.3 Formatting Map Visual

Options include:

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a. Map Styles

- Road (default)
- Aerial
- Dark
- Light

b. Zoom Controls

Enable zoom buttons for easier map navigation.

c. Data Colors

Customize color of bubbles

d. Size Controls

Adjust bubble scaling for clearer visibility.

e. Title and Formatting Controls

Same as other visuals.

5.2 Filled Map

5.2.1 Purpose

A filled map (choropleth map) shades entire geographic regions based on a quantitative value.

5.2.2 When to Use

- To compare regions at a high level
- To identify areas with no sales or low penetration
- To perform region-based analysis for strategy development

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5.2.3 Example Insights

If certain states display no color, it indicates:

- No sales in those states
- These regions are market opportunities
- Managers can develop sales strategies to improve penetration

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5.3 ArcGIS for Power BI

5.3.1 Features

- Rich mapping detail
- Multiple layers
- Heat maps
- Reference boundaries

5.3.2 Requirements

- Requires **internet**
- Works best when **latitude/longitude** are provided

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5.3.3 Using ArcGIS

Provide:

- **Latitude**
- **Longitude**
- **Size field** (e.g., Profit)

Bubble sizes change based on the value.

5.4 Azure Maps

Not demonstrated because:

- Requires **sign-in**
- Often restricted without an Azure account

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6. Gauge Chart

6.1 Purpose



Gauge charts resemble a speedometer and help display:

- **Target achievement**
- **Progress toward a goal**
- **KPI status**

Examples:

- Sales vs. Target
- Profit vs. Budget
- Production vs. Capacity

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6.2 Creating a Gauge Chart

1. Select the **Gauge** visual.
2. Provide:
    - **Value** → Sales (actual)
    - **Target value** → Target Sales
    - **Minimum and Maximum** must be manually defined

Power BI auto-assigns min/max values, but they should be overridden for clarity.

6.3 Setting Gauge Axis

Use the **Gauge Axis** settings:

- **Minimum value** (e.g., 25,000)
- **Maximum value** (e.g., 60,000)
- **Target value** (e.g., 45,000)

Gauge visually shows:

- Actual value (pointer)
- Target threshold (line)
- Range from minimum to maximum

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6.4 Formatting Gauge Chart

a. Data Labels

Customize:

- Value font size
- Color
- Boldness

b. Target Label

Shows the target value clearly (can be colored differently).

c. Gauge Appearance

- Change needle color
- Change range color
- Customize background

d. Title and Effects

- Modify title text
- Add border
- Change alignment and font size

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7. Summary

These visuals help analyze one quantitative variable and one or more categorical variables in different ways:

Pie Chart / Donut Chart

- Best for small category counts (≤8)
- Shows proportion

Tree Map

- Best for large category counts (>10)
- Shows proportional area

Map Visuals

- Show geographic distribution
- Combine quantitative and spatial data
- Provide strategic insights on regional performance

Gauge Chart

- Ideal for KPI tracking
- Shows progress toward a target

Each visual serves a distinct purpose, and Power BI's formatting and drill-down capabilities help convert raw data into intuitive insights.

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